

S.No. : 611

NCS 4404

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Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 43212 Roll
No.

B. Tech. Examination, 2026

(Even Semester)

COMPUTER ORGANIZATION & ARCHITECTURE

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION – A

1. Attempt all parts of the following : $8 \times 1 = 8$

- (a) List three types of control signals.
- (b) How objective Ma many types of buses are in computer architecture?
- (c) What is synchronous communication?

[P. T. O.

- (d) Differentiate interrupts and exceptions.
- (e) How many bits are there in the data and address input of the memory?
- (f) Explain two logical micro-operations. *AND OR*
- (g) Right circular shift of the following data :
- (i) 10101110
- (ii) 01100110
- (h) How many 128×8 memory chips are needed to provide a memory capacity 4096×16 ?

SECTION - B

2. Attempt any two parts of the following : *2x6=12*
What do mean Micro instruction sequence
- (a) Explain internal organization of $2M \times 8$ asynchronous DRAM chip.
Defn between MISC & SISC
- (b) How many switch points are there in a cross bar switch network that connects 'p' processors to 'm' memory modules?
- (c) Draw the flow chart and explain about booth's algorithm. Multiply 100111 with 11011 using booth's algorithm.

- (d) What is the difference between hardwired control design and a microprogrammed control design?

SECTION - C

Note :- Attempt all questions. Attempt any two parts from each questions. $8 \times 5 = 40$

3. (a) Explain interconnection structure and its types "in computer organization and architecture."
- (b) Draw and explain Von-Neumann architecture of a computer in detail.
- (c) Explain various cache write policies.
4. (a) *What is 2D & 2.5D memory Organization*
 What do you understand by Amdahl's law? *zad*
 Explain with suitable example.
- (b) What are instruction pipelines? Also explain the performance of pipeline.
- (c) Perform the division process of 00001111 by 0011 (use a dividend of 8 bits).
5. (a) What do you mean by interrupts? Explain different types of interrupts.

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- (b) Explain the mechanism involved in magnetic disk and magnetic tapes.
 - (c) Explain any five addressing modes with examples.
6. (a) Explain difference between synchronous and asynchronous communication along with standard communication interfaces.
- (b) What is the difference between I/O interface and I/O port.
- (c) Define briefly the bus arbitration technique.
